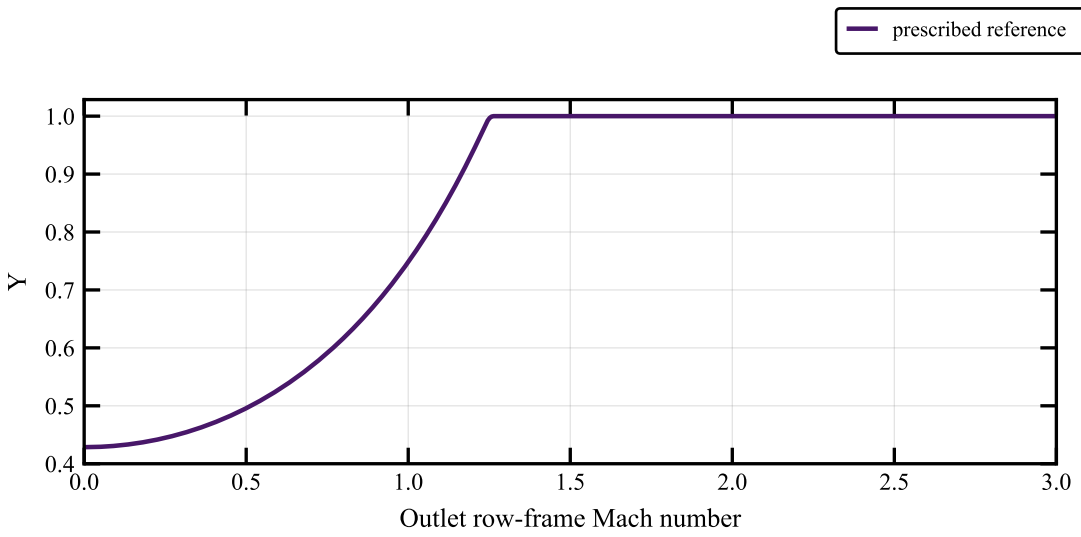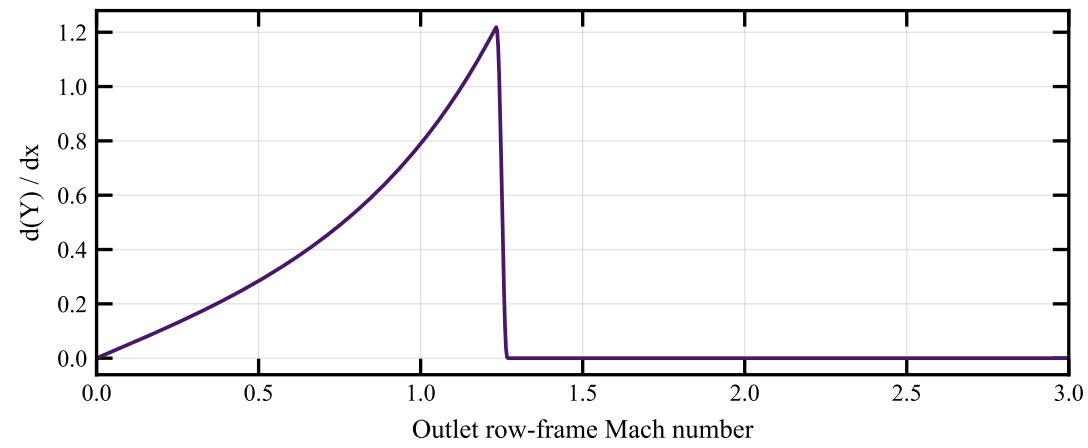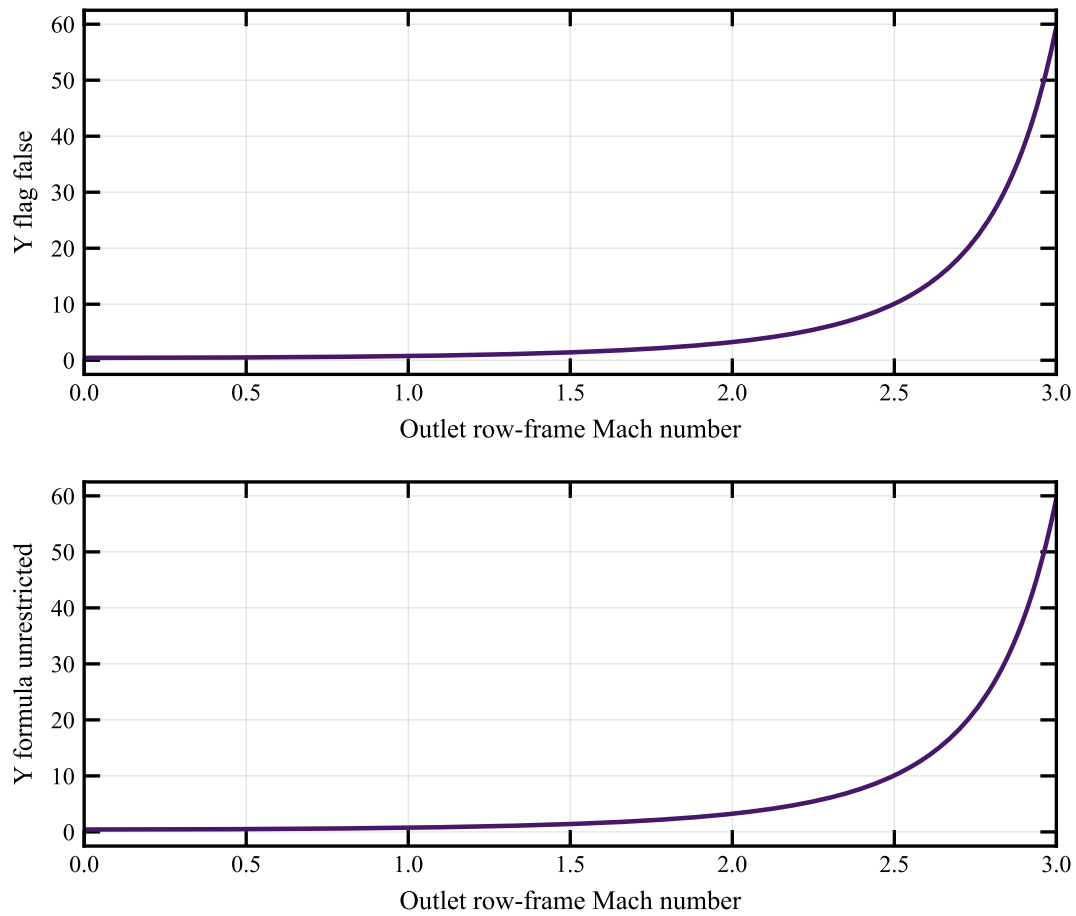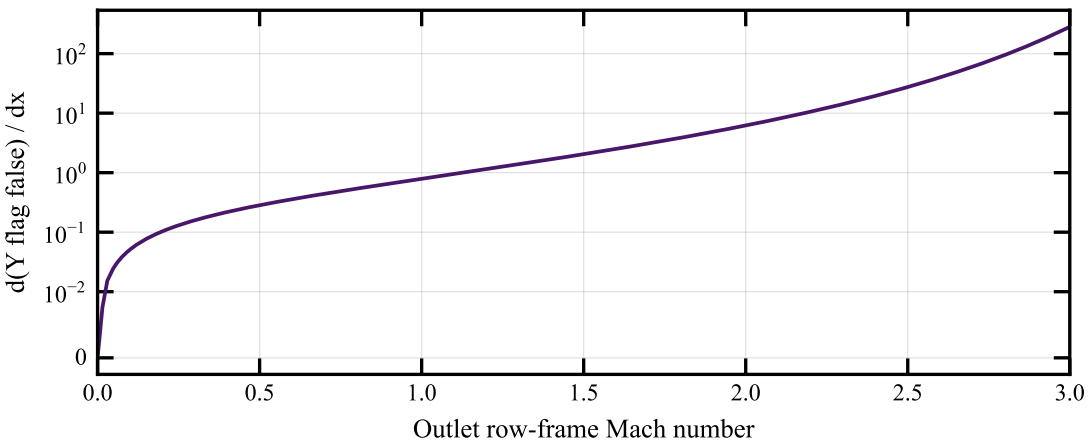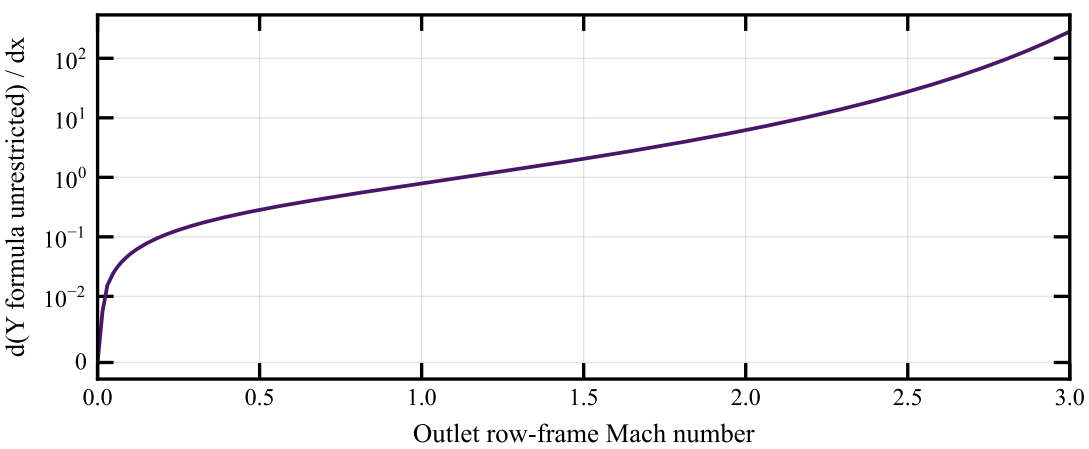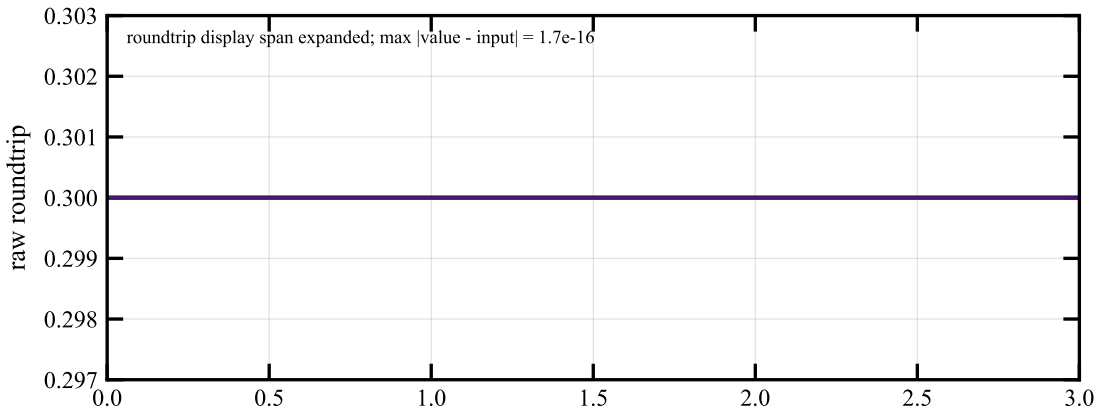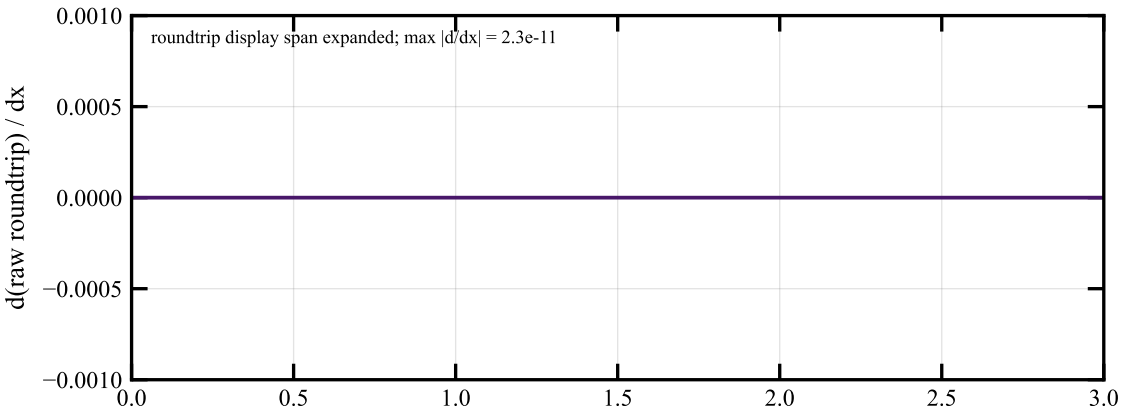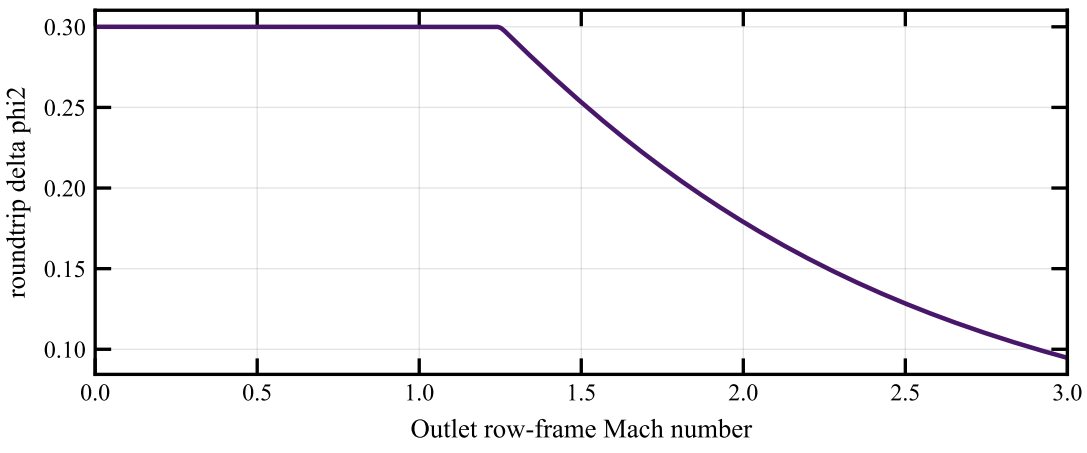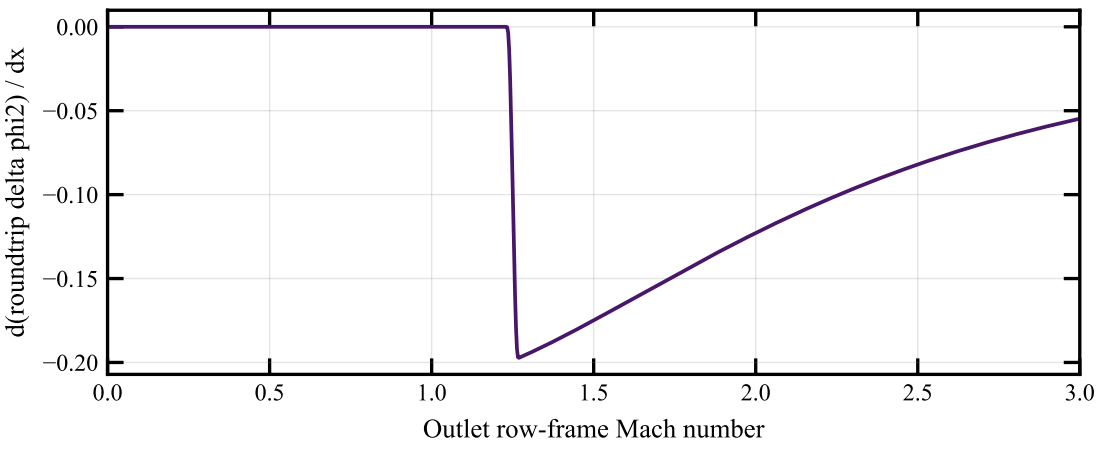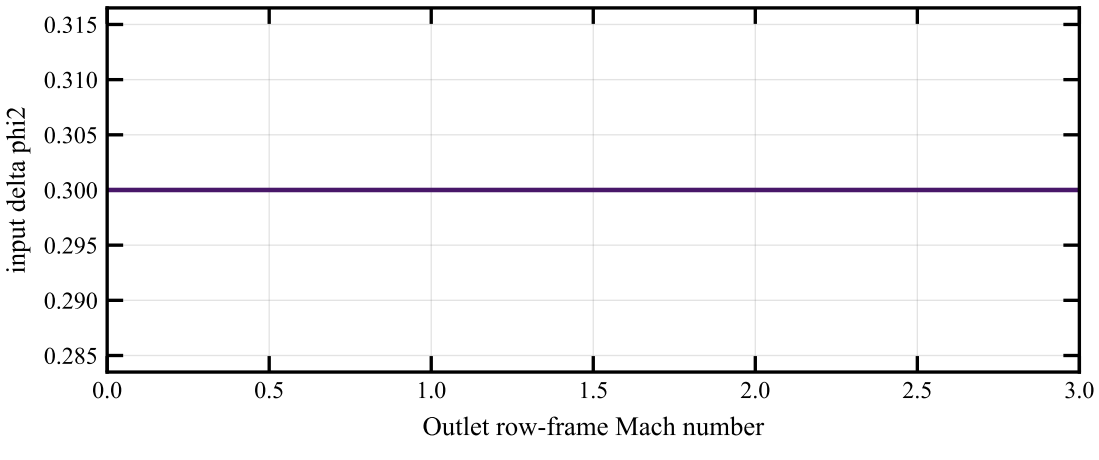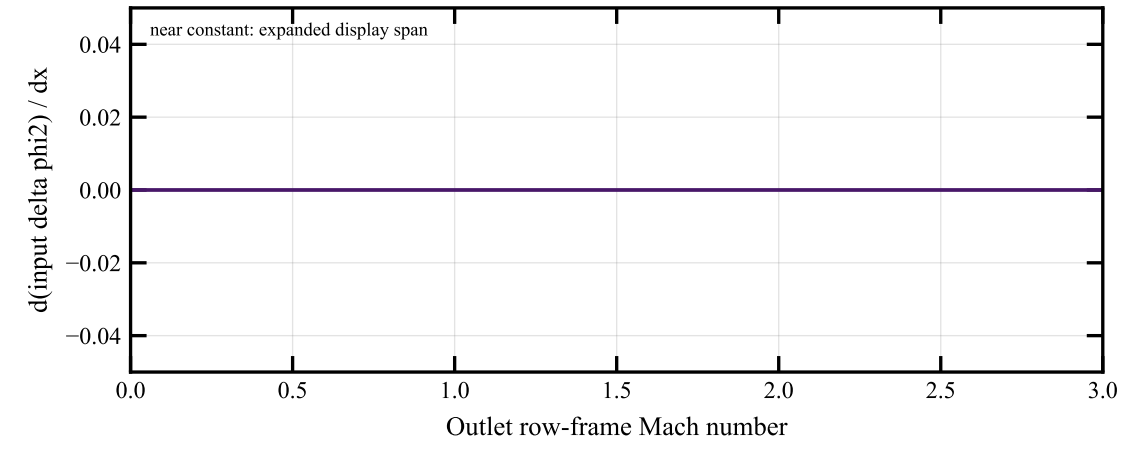conversion mach d0.3 — values and first derivatives